

Question 35 (3 marks)

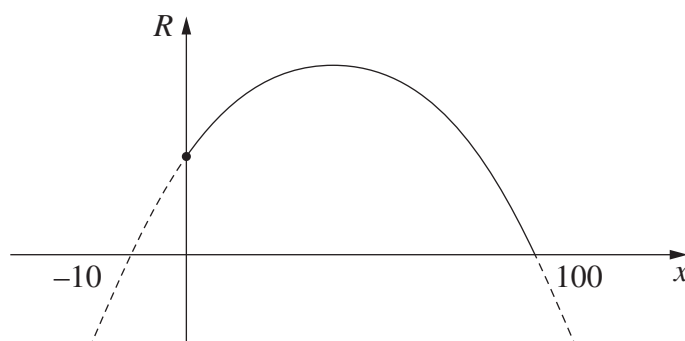
A publisher sells a book for \$10. At this price, 5000 copies of the book will be sold and the revenue raised will be $5000 \times 10 = \$50\,000$.

The publisher is considering increasing the price of the book. For every dollar the price of the book is increased, the publisher will sell 50 fewer copies of the book.

If the publisher charges $(10 + x)$ dollars for each book, a quadratic model for the revenue raised, R , from selling the books is

$$R = -50x^2 + 4500x + 50\,000.$$

A graph of this quadratic model for revenue is shown. A dashed line is used for values of x which are not relevant to the practical context of this problem.



- (a) By first finding a suitable value of x , find the price the publisher should charge for each book to maximise the revenue raised from sales of the book.

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- (b) Find the value of the intercept of the parabola with the vertical axis.

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Do NOT write in this area.